Wildlife Threat Prediction

A Machine Learning Model for Threatened Species

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Introduction

What the project is and why we need it

Why It Matters

Conservation teams cannot manually review every species. A predictive model helps prioritize attention where it is needed most.

Simple Idea

Think of it as a first filter. It does not replace experts, but helps surface species that deserve a closer look.

What It Does

The model learns patterns from species data, then flags those with a higher predicted risk for expert review.

Data

Two datasets, their size, and how we merged them

Dataset 1 — WWF

82 rows

Used to define threatened status.

Dataset 2 — IUCN

159,542 rows

Main table with taxonomy and species context.

How we merged them

We matched records using scientific names, then built one final modelling table from the combined data.

After the Merge

From splitting to feature engineering to model selection

1

Split

Divided data into train and test sets to check generalization.

2

Encode

Converted categories into numbers. Machine learning needs numeric input.

3

Engineer

Added useful features like taxo_risk to help the model see combined patterns.

4

Select

Compared models and chose the strongest one for deployment.

Goal: Turn raw species data into a clean training set and choose the model with the best balance of performance and stability.

Models We Trained

The scores and the model we selected

Model	Test Accuracy	Status
Decision Tree	74.4%	—
Bagging	74.5%	✓ Selected
AdaBoost	74.4%	—
Gradient Boosting	74.4%	—

Selected: Bagging had the highest test accuracy and the most stable validation results.

Boosting & Improvement Attempts

How we tried to make the model better

What We Tried

- Balanced class weights
- Hyperparameter tuning
- Feature engineering with taxo_risk
- AdaBoost and Gradient Boosting testing

Metrics Tracked

- Accuracy
- F1 score
- Recall for the threatened class
- Cross-validation stability

Main Lesson: Even after improvement attempts, the model still had weak recall for threatened species — accuracy alone was not enough.

Predictions

Top 5 predicted threatened species from the final model

- ✓ ***Acipenser ruthenus*** — Fish (Sturgeon) — *Correctly predicted*
- ✓ ***Cookeconcha contorta*** — Land Snail — *Correctly predicted*
- ✓ ***Acipenser oxyrinchus*** — Fish (Sturgeon) — *Correctly predicted*
- ✓ ***Acipenser transmontanus*** — Fish (Sturgeon) — *Correctly predicted*
- ! ***Meta stridulans*** — Spider (false positive) — *False positive — overestimated risk*

4
correctly predicted

1
false positive

Results

What the scores mean

74.5%

Final Test Accuracy

0.95

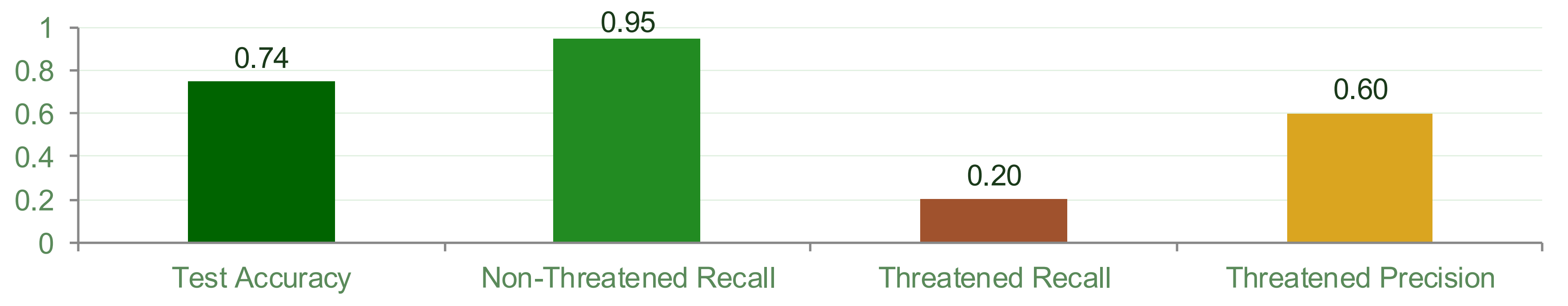
Recall — Non-Threatened

0.20

Recall — Threatened

0.60

Precision — Threatened



Next Steps

How to make the screening more robust

01

Add More Data

Include additional ecological and geographic data for richer feature coverage.

02

Better Engineering

Apply more advanced feature engineering to expose hidden patterns.

03

Imbalance Methods

Try improved techniques to address class imbalance in the threatened label.

04

Tune Threshold

Adjust classification threshold to improve recall for threatened species.

The model works as an initial screen, but would become much more robust with extra data, better techniques, and more refinement.

Thank You

" The future of our species depends on the choices we make now. "

— Sir David Attenborough